# AI-Assisted Literature Review Lab

Using Hugging Face + Google Colab

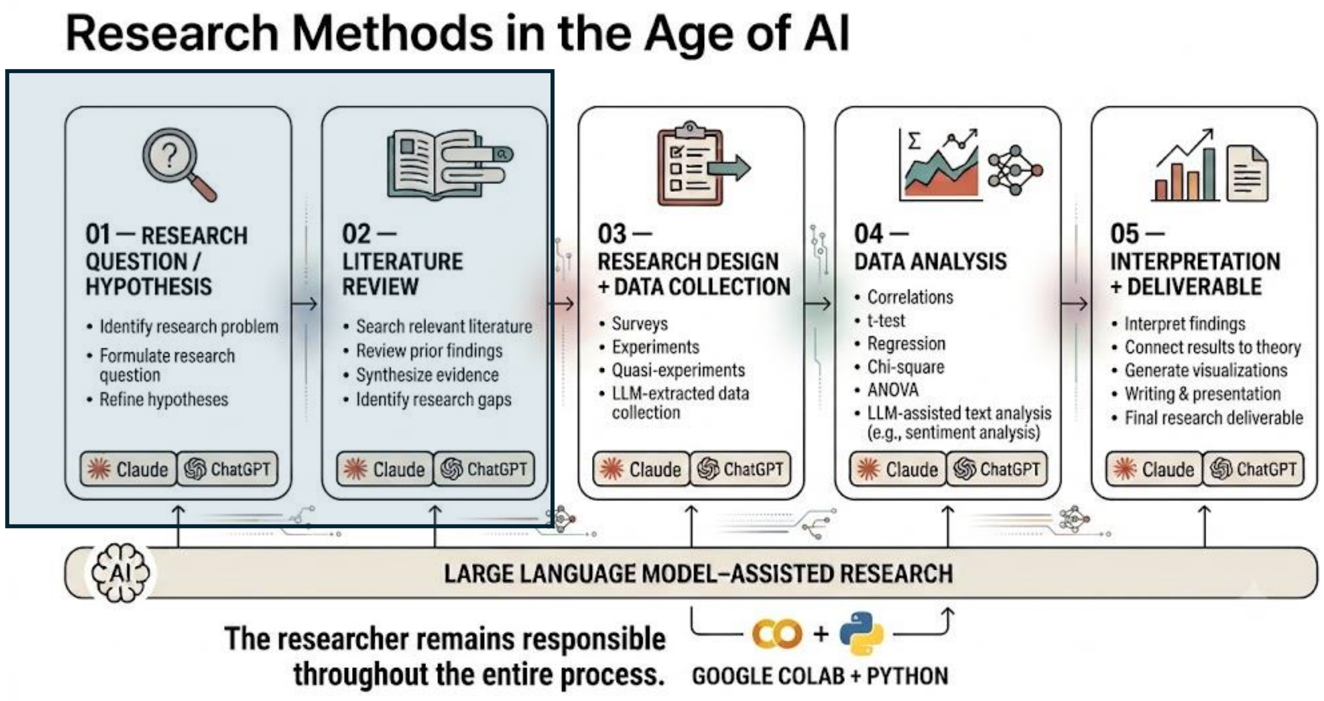
# Today’s Lab

- We will use an existing AI model to help **screen research abstracts**.
- By the end, you will:
  - Use AI to screen abstracts
  - Compare **AI vs. human judgment**
  - Identify where AI gets it wrong
- AI helps. The researcher decides.

# Today’s Workflow

- Work with your group’s abstract dataset
- Load the CSV file into Google Colab
- Use an existing LLM model with Hugging Face
- Compare your judgment with the model’s judgment
- Change the labels and try again
- Discuss one useful disagreement with the class

# Where Are We in the Research Process?



# Why Not Just Use ChatGPT?

ChatGPT is useful for one-off thinking, brainstorming, and explanation.

- For research workflows, we often need code because it can:
- process many abstracts the same way
- save the model outputs
- make the process easier to check and repeat
- help us compare human and model judgments systematically

# 1. Your Group Dataset

Each group should have:

- One research question
- 3 peer-reviewed article abstracts per person
- About 12–15 abstracts total for a group of 4–5
- One combined CSV file with the columns shown on the right
- Use abstracts from peer-reviewed journal articles only

| Group Member | abstract |
|---|---|
| Max | This study examines how journalists use generative AI in daily work... |
| Alex | Using survey data, this study explores how young adults encounter news... |
| Sarah | This article analyzes how platform users evaluate health misinformation... |

# 2. Load Your Abstracts Into Colab

Python

Run

```python
import pandas as pd

df = pd.read_csv("Group##_abstracts.csv")
df.head()
```

Then:

Python

Run

```python
abstracts = df["abstract"].tolist()
```

# 3. Human Judgment Comes First

- Before using the model, each person should return to the 3 abstracts they contributed.
- Read each abstract yourself first.
- Record your private judgment before looking at the model output.
- Use one of these labels:
    - Clearly relevant
    - Possibly relevant
    - Probably unrelated

# 4. Define the Task

- Tell the Model What We Are Looking For
- The model does not independently know what “relevant” means.
- Your labels tell the model what kinds of evidence to look for.
- Example research question: “How is generative AI changing journalists’ professional practices?”

Python

```python
labels = [
    "clearly relevant",
    "possibly relevant",
    "probably unrelated"
]
```

# 5. Load the Pretrained Model

Python

Run

```python
from transformers import pipeline

classifier = pipeline(
    "zero-shot-classification",
    model="INSTRUCTOR_SELECTED_MODEL"
)
```

# 6. Try One Abstract First

Python

Run

```python
result = classifier(
    abstracts[0],
    candidate_labels=labels
)

result
```

# Do you agree or disagree with the model?

# 8. Change the Labels

Use labels that fit your research question, then compare whether the model judgment improves.

Python

```python
labels = [
    "generative AI in journalistic work",
    "AI in another professional context",
    "journalism without generative AI"
]
```

# 9. Process the Full Abstract Set

Python

Run

```python
for abstract in abstracts:
    result = classifier(
        abstract,
        candidate_labels=labels
    )

    print(result["labels"][0])
```

# 10. Save the Model Results

Python

Run

```python
results = []

for abstract in abstracts:
    result = classifier(
        abstract,
        candidate_labels=labels
    )
    results.append(result)
```

Then:

Python

Run

```python
results[0]
```

# 11. Compare Human + Model Judgment

- Review the abstracts **you personally contributed**
- Compare your human judgment with the **AI judgment**
- For each abstract, record **Agree with AI: Yes or No**
- If you **agree**, briefly explain what evidence in the abstract supports the AI judgment
- If you **disagree**, briefly explain what the model may have misunderstood or missed
- If you are **unsure**, discuss the abstract with your group before making **your final Yes/No decision**
- You are responsible for completing the rows for **your own abstracts**

| Group Member | Abstract | Human Judgment | AI Judgment | Agree with AI? | Reason |
|---|---|---|---|---|---|
| Alex | … | Clearly relevant | Relevant | Yes | The abstract directly examines… |
| Maya | … | Possibly relevant | Not relevant | No | The model may have missed… |

# Assignment 2 Deliverable

- Upload one agreement table file on Blackboard.
- One file per group.
- Due by 5 PM tomorrow.
- Your file should include the columns shown below.

| Group Member | Abstract | Human Judgment | AI Judgment | Agree with AI? | Reason |
|---|---|---|---|---|---|
| Alex | … | Clearly relevant | Relevant | Yes | The abstract directly examines… |
| Maya | … | Possibly relevant | Not relevant | No | The model may have missed… |

# Why Might You and AI Disagree?

- The abstract is **unclear**
- The labels are **not specific enough**
- The abstract lacks information
- AI focuses on the **wrong words**
- People interpret the RQ differently

# Recap

- AI can help us screen many abstracts.
- Python helps us repeat the same process.
- AI can make mistakes.
- Researchers make the final decision.

# Coming Next: Block 1 Knowledge Assessment

- Week 5 Day 2
- Open-book, paper-based assessment
- You may discuss with 1–2 classmates before submitting your own answers
- Focus areas:
  - Research methods in the age of AI
  - Basic Transformer concepts and limitations
  - Basic Python code reading and interpretation
  - Critical thinking about AI-assisted research workflows